MILITARY SPECIFICATION

TRANSISTOR, PNP, SILICON, DOUBLE-EMITTER

TYPE 3N108

1. SCOPE

1.1 Description.- This specification covers the detail requirements for a silicon, PNP, double-emitter transistor, characterized for low-level, chopper applications, and is in accordance with MIL-S-19500, except as otherwise specified herein.

1.2 Mechanical dimension and outline.- TO-72 (See figure 1).

1.3 Absolute maximum ratings.-

P_T 1/ T_C = 25°C	P_T 2/ T_A = 25°C	T_{stg}	V_{E_1C} or V_{E_2C}	V_{E_1B} or V_{E_2B}	$V_{E_1E_2}$	V_{CB}
mW	mW	°C	Vdc	Vdc	Vdc	Vdc
600	300	-65 to +200	-50	-50	±50	-50

I_C	I_B	I_{E_1} or I_{E_2}
mAdc	mAdc	mAdc
±20	±20	±10

1/ Derate linearly at 3.43 mW/°C above T_C = 25°C.
2/ Derate linearly at 1.71 mW/°C above T_A = 25°C.

1.4 Primary electrical characteristics.-

	$\lvert V_{E_1E_2} \rvert$	$r_{e_1e_2(on)}$	h_{fe}
	I_B = -1 mAdc $I_{E_1} = I_{E_2}$ = 0 (See figure 2)	I_B = -1 mAdc $I_{E_1} = I_{E_2}$ = 0 I_{e_1} = 100 μAac f = 1 kc	V_{CE_1} (or V_{CE_2}) = -6 Vdc I_C = -1 mAdc I_{E_2} (or I_{E_1}) = 0 f = 4 Mc
	μVdc	ohms	---
Min.	---	10	3.0
Max.	30	50	---

FSC 5961

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2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARDS

MILITARY

MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions.- The abbreviations, symbols, and definitions used herein are defined in MIL-S-19500, and as follows:

$I_{E_1E_2}$ . . . . . Emitter to emitter cutoff current, collector short-circuited to base and open circuited to emitters.

$V_{E_1E_2}$ . . . . . Emitter one to emitter two voltage (d.c.).

$|V_{E_1E_2}|$ . . . . . Magnitude of emitter to emitter offset voltage (d.c.).

BV_{ECO} . . . . . Breakdown voltage, emitter to collector, base open.

$BV_{E_1E_2}$ . . . . . Breakdown voltage, emitter one to emitter two, collector short-circuited to base and open circuited to emitters.

$|\Delta V_{E_1E_2}|_{I_B}$ . . . . . Magnitude of algebraic difference between offset voltages measured at two different base currents.

$|\Delta V_{E_1E_2}|_{T_A}$ . . . . . Magnitude of algebraic difference between offset voltages measured at two different temperatures.

$r_{ele2}(on)$ . . . . . Dynamic "on" series impedance between emitter one and emitter two.

C_{ibo} . . . . . Input capacitance, (common base), output open circuited.

I_e . . . . . Emitter current (r.m.s.).

Offset voltage . . The emitter to emitter offset voltage is defined as the open circuit voltage appearing between the two emitters when the collector-base junction is forward biased.

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3.3 Design, construction and physical dimensions.- Transistors shall be of the design, construction, and physical dimensions shown on figure 1.

3.4 Performance characteristics.- Performance characteristics shall be as specified in 4.3 and 4.4.

3.5 Marking.- The following marking specified in MIL-S-19500 may be omitted from the body of the transistor:

a. Country of origin.

b. Manufacturer's identification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Qualification tests.- Qualification tests shall be conducted at a a laboratory satisfactory to the Naval Ship Engineering Center. Qualification tests shall consist of the tests specified in 4.3 and 4.4. (Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (See 6.1).)

4.2 Quality conformance inspection.- Quality conformance inspection shall consist of the examinations and tests specified in 4.3 and 4.4.

4.3 Group A inspection.- Group A inspection shall consist of the examinations and tests specified in table I.

4.4 Group B inspection.- Group B inspection shall consist of the examinations and tests specified in table II.

4.5 Methods of examination and test.- Methods of examination and test shall be as specified in tables I and II, and as follows:

4.5.1 Shock.- The shock testing apparatus shall be capable of providing shock pulses of the specified peak acceleration, waveform, and pulse duration to the body of the device. The acceleration pulse, as determined from the unfiltered output of a transducer with a natural frequency greater than 10,000 cycles per second shall be a half sine waveform with an allowable distortion not greater than ±20 percent of the specified peak acceleration. The pulse duration shall be measured between the points at 10 percent of the peak acceleration during rise and at 10 percent of the peak acceleration during decay. Absolute tolerance of the pulse duration shall be ±40 percent of the specified duration.

4.5.2 Open-circuit input capacitance.- Method of test shall be in accordance with Method 3240 of MIL-STD-750 except that the output capacitor shall be omitted, and the alternate emitter shall be open circuited.

4.5.3 Emitter to emitter cutoff current.- The transistor shall be tested in the circuit shown on figure 4. The specified voltage shall be applied between the two emitter terminals. The current following into emitter-one shall then be measured.

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4.5.4 Emitter to collector breakdown voltage.- Method of test shall be in accordance with method 3011 of MIL-STD-750, Condition D except that the collector and the specified emitter terminal shall be interchanged and the alternate emitter shall be open circuited.

4.5.5 Emitter to emitter breakdown voltage.- The transistor shall be tested in the circuit shown on figure 4. The resistor R_1 is a current limiting resistor and should be of sufficiently high resistance to avoid excessive current flowing through the transistor and the current meter. The voltage shall be gradually increased from zero until either the minimum limit of $BV_{E_1E_2}$ or the specified test current is reached. The transistor is acceptable if the minimum limit for $BV_{E_1E_2}$ is reached before the test current reaches the specified value. If the specified test current is reached first, the device is rejected.

4.5.6 Emitter to emitter offset voltage.- The transistor shall be tested in the circuit shown on figure 2. The base current shall be adjusted to the specified value. The voltage between the two emitters shall then be measured using a voltmeter with an input impedance high enough that halving it does not change the measured value within the required accuracy of the measurement.

4.5.7 Emitter to emitter offset voltage change with base current.- The offset voltage and its polarity shall be measured as specified in 4.5.6 at the two specified base currents. The magnitude of the algebraic difference between these two values shall then be computed.

4.5.8 Emitter to emitter offset voltage change with temperature.- The offset voltage and its polarity shall be measured as specified in 4.5.6 at the two specified temperatures. The magnitude of the algebraic difference between these two values shall then be computed.

4.5.9 Dynamic "on" series impedance.- The transistor shall be tested in the circuit shown on figure 3. The base current shall be adjusted to the specified value and an a.c. sinusoidal signal current of the specified rms value applied through both emitters. The rms voltage, $V_{e_1e_2}$, between the two emitters shall be measured using an a.c. voltmeter with an input impedance high enough that halving it does not change the measured value within the required accuracy of the measurement. The dynamic "on" series impedance shall then be determined as follows:

$$r_{e_1e_2}(on) = \frac{V_{e_1e_2}}{I_{e_1}}$$

where $V_{e_1e_2}$ is the rms voltage between the two emitters.

5. PREPARATION FOR DELIVERY

5.1 See MIL-S-19500.

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6. NOTES

6.1 The activity responsible for the qualified products list is the Naval Ship Engineering Center, Department of the Navy, Washington, D. C. 20360, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification". (Copies of "Provisions Governing Qualification" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, Pa. 19120.)

Review activities:
Navy - SH, EC

User activities:
Navy - AS, OS, MC, CG

Preparing activity:
Navy - SH
(Project 5961-N038(NAVY))

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Table I - Group A Inspection.

EXAMINATION OR TEST	MIL-STD-750		LTPD	SYMBOL	LIMITS		UNIT
	METHOD	CONDITIONS			MIN.	MAX.	
Subgroup 1			10				
Visual and mechanical examination	2071						
Subgroup 2			5				
Collector to base cutoff current	3036 Cond. D	V_{CB} = -30 Vdc $I_{E_1} = I_{E_2} = 0$		I_{CBO}	---	-0.25	nAdc
Emitter to base cutoff current	3061 Cond. D	V_{E_1B} = -25 Vdc $I_C = 0$ $I_{E_2} = 0$		I_{E_1BO}	---	-0.1	nAdc
Emitter to base cutoff current	3061 Cond. D	V_{E_2B} = -25 Vdc $I_C = 0$ $I_{E_1} = 0$		I_{E_2BO}	---	-0.1	nAdc
Emitter to emitter cutoff current (See 4.5.3)	----	$V_{E_1E_2}$ = +25 Vdc $V_{CB} = 0$		$I_{E_1E_2}$	---	+0.1	nAdc
Emitter to emitter cutoff current (See 4.5.3)	----	$V_{E_1E_2}$ = -25 Vdc $V_{CB} = 0$		$I_{E_1E_2}$	---	-0.1	nAdc
Collector to base breakdown voltage	3001 Cond. D	I_C = -1.0 µAdc $I_{E_1} = I_{E_2} = 0$		BV_{CBO}	-50	---	Vdc
Emitter to collector breakdown voltage (See 4.5.4)	----	I_{E_1} = -1.0 µAdc $I_{E_2} = 0$ $I_B = 0$		BV_{E_1CO}	-50	---	Vdc
Emitter to collector breakdown voltage (See 4.5.4)	----	I_{E_2} = -1.0 µAdc $I_{E_1} = 0$ $I_B = 0$		BV_{E_2CO}	-50	---	Vdc
Emitter to base breakdown voltage	3026 Cond. D	I_{E_1} = -1.0 µAdc $I_{E_2} = 0$ $I_C = 0$		BV_{E_1BO}	-50	---	Vdc

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Table I - Group A Inspection (Cont'd).

EXAMINATION OR TEST	MIL-STD-750 METHOD	MIL-STD-750 CONDITIONS	LTPD	SYMBOL	LIMITS MIN.	LIMITS MAX.	UNIT
Subgroup 2 (Cont'd)							
nitter to base breakdown voltage	3026 Cond. D	I_{E_2} = -1.0 µAdc I_{E_1} = 0 I_C = 0		BV_{E_2BO}	-50	---	Vdc
nitter to emitter breakdown voltage (See 4.5.5)	----	I_{E_1} = +1.0 µAdc V_{CB} = 0		$BV_{E_1E_2}$	+50	---	Vdc
itter to emitter breakdown voltage (See 4.5.5)	----	I_{E_1} = -1.0 µAdc V_{CB} = 0		$BV_{E_1E_2}$	-50	---	Vdc
Subgroup 3			5				
gnitude of emitter to emitter offset voltage (See 4.5.6)	----	I_B = -1.0 mAdc I_{E_1} = I_{E_2} = 0 T_A = 25°C		$\lvert V_{E_1E_2} \rvert$	---	30	µVdc
gnitude of emitter to emitter offset voltage (See 4.5.6)	----	I_B = -1.0 mAdc I_{E_1} = I_{E_2} = 0 T_A = -25°C		$\lvert V_{E_1E_2} \rvert$	---	30	µVdc
gnitude of emitter to emitter offset voltage (See 4.5.6)	----	I_B = -1.0 mAdc I_{E_1} = I_{E_2} = 0 T_A = +100°C		$\lvert V_{E_1E_2} \rvert$	---	30	µVdc
gnitude of emitter to emitter offset voltage change with base current (See 4.5.7)	----	$I_B(1)$ = -1.5 mAdc $I_B(2)$ = -0.5 mAdc I_{E_1} = I_{E_2} = 0		$\lvert \Delta V_{E_1E_2} \rvert_{I_B}$	---	20	µVdc
gnitude of emitter to emitter offset voltage change with temperature (See 4.5.8)	----	I_{B_1} = -1.0 mAdc I_{E_1} = I_{E_2} = 0 $T_A(1)$ = 100°C $T_A(2)$ = -25°C		$\lvert \Delta V_{E_1E_2} \rvert_{T_A}$	---	50	µVdc
namic "on" series impedance See 4.5.9)	----	I_{B_1} = -1.0 mAdc I_{E_1} = I_{E_2} = 0 I_{e_1} = 100 µAac rms f = 1 kc		$r_{e_1e_2(on)}$	10	50	ohms

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Table I - Group A Inspection (Cont'd).

EXAMINATION OR TEST	MIL-STD-750 METHOD	MIL-STD-750 CONDITIONS	LTPD	SYMBOL	LIMITS MIN.	LIMITS MAX.	UNIT
Subgroup 4			10				
Small-signal short-circuit forward current transfer ratio	3306	V_{CE_1} = -6.0 Vdc I_C = -1.0 mAdc I_{E_2} = 0 f = 4.0 Mc		h_{fe_1}	3.0	---	---
Small-signal short-circuit forward current transfer ratio	3306	V_{CE_2} = -6.0 Vdc I_C = -1.0 mAdc I_{E_1} = 0 f = 4.0 Mc		h_{fe_2}	3.0	---	---
Open-circuit output capacitance	3236	V_{CB} = -6.0 Vdc $I_{E_1} = I_{E_2}$ = 0 100 kc $\leq$ f $\leq$ 1 Mc		C_{obo}	---	10	pf
Open-circuit input capacitance (See 4.5.2)	3240	V_{E_1B} = -6.0 Vdc I_C = 0 I_{E_2} = 0 100 kc $\leq$ f $\leq$ 1 Mc		C_{ibo_1}	---	3.0	pf
Open-circuit input capacitance (See 4.5.2)	3240	V_{E_2B} = -6.0 Vdc I_C = 0 I_{E_1} = 0 100 kc $\leq$ f $\leq$ 1 Mc		C_{ibo_2}	---	3.0	pf
Emitter to emitter cutoff current (See 4.5.3)	----	$V_{E_1E_2}$ = +25 Vdc V_{CB} = 0 T_A = 100°C		$I_{E_1E_2}$	---	+10	nAdc
Emitter to emitter cutoff current (See 4.5.3)	----	$V_{E_1E_2}$ = -25 Vdc V_{CB} = 0 T_A = 100 °C		$I_{E_1E_2}$	---	-10	nAdc

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Table II - Group B Inspection,

EXAMINATION OR TEST	MIL-STD-750		LTPD	SYMBOL	LIMITS		UNIT
	METHOD	CONDITIONS			MIN.	MAX.	
Subgroup 1			20				
Physical dimensions	2066	See figure 1					
Subgroup 2			15				
Solderability	2026						
Thermal shock (temperature cycling)	1051 Cond. C						
Thermal shock (glass strain)	1056 Cond. A						
Moisture resistance	1021						
Subgroup 3			15				
Shock (See 4.5.1)	2016	1500 G, 0.5 msec 5 blows each orientation: X_1, Y_1, Y_2, Z_1					
Vibration fatigue	2046	nonoperating					
Vibration, variable frequency	2056						
Constant acceleration (centrifuge)	2006	20,000 G each orientation: X_1, Y_1, Y_2, Z_1					
Subgroup 4			10				
Terminal strength (lead fatigue)	2036 Cond. E						

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Table II - Group B Inspection (Cont'd).

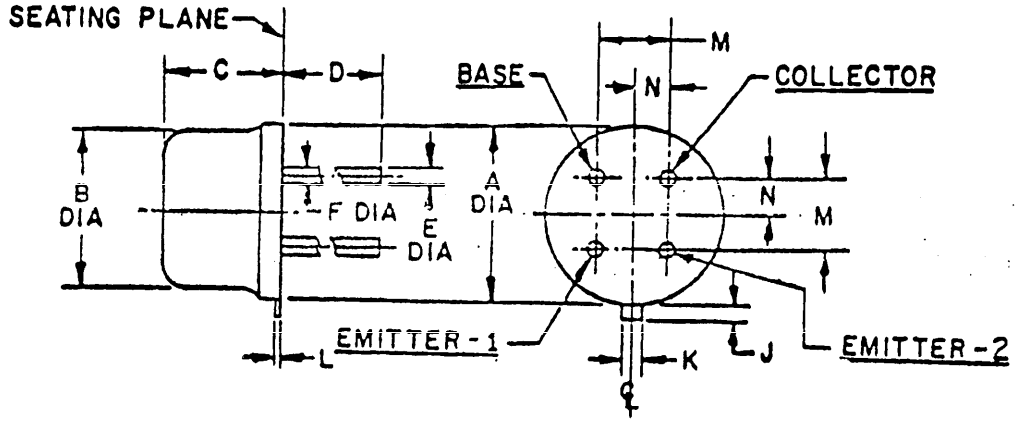
EXAMINATION OR TEST	MIL-STD-750		LTPD	SYMBOL	LIMITS		UNIT
	METHOD	CONDITIONS			MIN.	MAX.	
Subgroup 4 (Cont'd)							
End point:							
Seal	MIL-STD-202 Method 112	Test cond. C Procedure III Test cond. A for gross leaks		leak rate	---	$5x10^{-7}$	atm cc per sec
Subgroup 5			10				
Salt atmosphere (corrosion)	1041						
Subgroup 6			$\lambda = 10$				
High-temperature life (nonoperating)	1031	T_A = 200°C					
Subgroup 7			$\lambda = 10$				
Steady-state operation life	1026	V_{CE} = 15 Vdc $V_{E_1E_2}$ = 0 P_T = 300 mW T_A = 25°C					
End points for subgroups 2 and 3:							
Magnitude of emitter to emitter offset voltage (See 4.5.6)		I_{B_1} = -1.0 mAdc $I_{E_1} = I_{E_2}$ = 0		$\lvert V_{E_1E_2} \rvert$	---	45	µVdc
Emitter to base cutoff current	3061 Cond. D	V_{E_1B} = -25 Vdc I_C = 0 I_{E_2} = 0		I_{E_1BO}	---	-0.5	nAdc
Emitter to base cutoff current	3061 Cond. D	V_{E_2B} = -25 Vdc I_C = 0 I_{E_1} = 0		I_{E_2BO}	---	-0.5	nAdc

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Table II - Group B Inspection (Cont'd).

EXAMINATION OR TEST	MIL-STD-750		LTPD	SYMBOL	LIMITS		UNIT
	METHOD	CONDITIONS			MIN.	MAX.	
End points for subgroups 6 and 7:							
Magnitude of emitter to emitter offset voltage (See 4.5.6)		I_{B_1} = -1.0 mAdc $I_{E_1} = I_{E_2} = 0$		$\lvert V_{E_1E_2} \rvert$	---	45	µVdc
Emitter to base cutoff current	3061 Cond. D	V_{E_1B} = -25 Vdc $I_C = 0$ $I_{E_2} = 0$		I_{E_1BO}	---	-0.5	nAdc
Emitter to base cutoff current	3061 Cond. D	V_{E_2B} = -25 Vdc $I_C = 0$ $I_{E_1} = 0$		I_{E_2BO}	---	-0.5	nAdc
Emitter to emitter breakdown voltage (See 4.5.5)		I_{E_1} = +1.0 µAdc $V_{CB} = 0$		$BV_{E_1E_2}$	+48	---	Vdc
Emitter to emitter breakdown voltage (See 4.5.5)		I_{E_1} = -1.0 µAdc $V_{CB} = 0$		$BV_{E_1E_2}$	-48	---	Vdc
Dynamic "on" series impedance (See 4.5.9)		I_{B_1} = -1.0 mAdc $I_{E_1} = I_{E_2} = 0$ I_{e_1} = 100 µAac rms f = 1 kc		$r_{e_1e_2(on)}$	---	60	ohms

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LTR	DIMENSIONS				NOTES
	INCHES		MILLIMETERS		
	MIN	MAX	MIN	MAX	
A	.209	.230	5.31	5.84	
B	.178	.195	4.52	4.95	
C	.170	.210	4.32	5.33	
D	.500	--	12.70	--	7
E	--	.021	--	.53	2,7
F	.016	.019	.41	.48	3,7
J	.028	.048	.71	1.22	6
K	.036	.046	.91	1.17	
L	—	.020	--	.51	
M	.0707 Nom		1.80 Nom		4
N	.0354 Nom		0.90 Nom		4

NOTES:

1. Metric equivalents in table and parentheses are shown for general information only and are based upon 1 inch = 25.4 mm.
2. Measured in the zone beyond 0.250 (6.35) from the seating plane.
3. Measured in the zone 0.050 (1.27) and 0.250 (6.35) from the seating plane.
4. When measured in a gaging plane $0.054\ ^{+.001}_{-.000}$ ($1.37\ ^{+.03}_{-.00}$) below the seating plane of the transistor, maximum diameter leads shall be within 0.007 (0.18) of their true location relative to a maximum width tab. Smaller diameter leads shall fall within the outline of the maximum diameter lead tolerance.
5. The collector shall be internally connected to the case.
6. Measured from the maximum diameter of the actual device.
7. All four leads.

Figure 1 - Physical dimensions and outline of transistor type 3N108.

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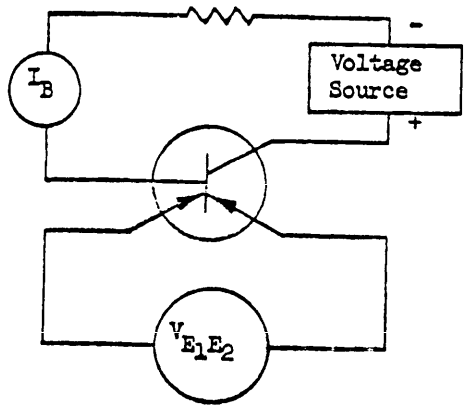


Figure 2 - Offset voltage test circuit.

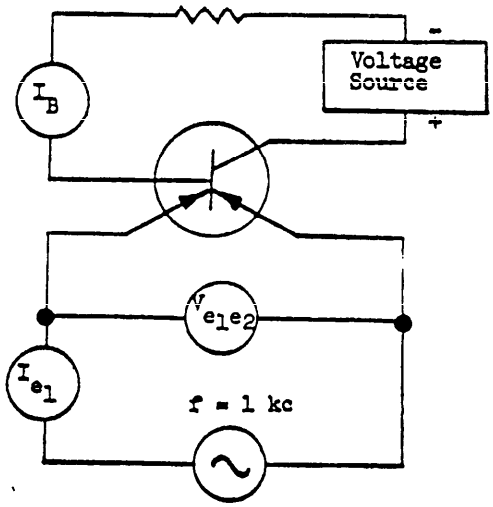


Figure 3 - Dynamic "on" series impedance test circuit.

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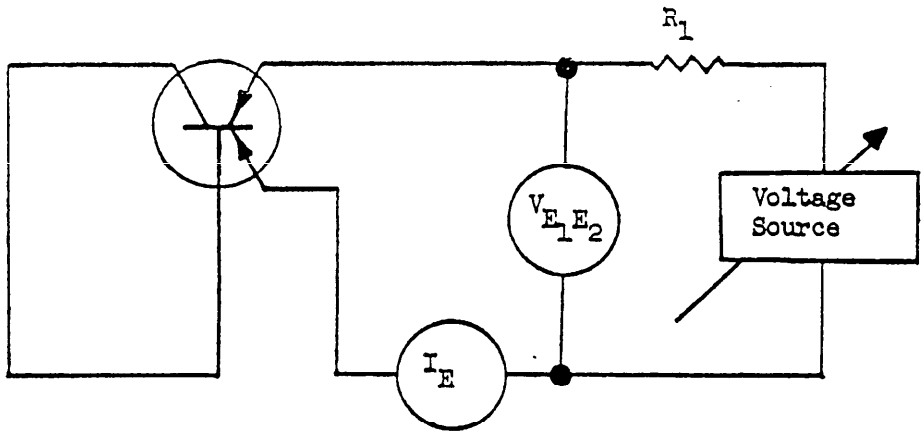


Figure 4 - Emitter to emitter cutoff current and Emitter to emitter breakdown voltage test circuit.

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